Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2026.DOI

Temporal Pixel Masking for Mitigating OCR-Based Text Recovery from Short-Exposure Screen Photography: A Feasibility Study with a UVC Camera

ZHEYUAN HUANG[1], CHEN XING[1]

[1]School of Computer Science and Technology, Zhejiang University of Water Resources and Electric Power, Hangzhou 310018, People's Republic of China

Corresponding author: Chen Xing (e-mail: xingch@zuwe.edu.cn).

This manuscript includes a minimal-risk human-subject usability study. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type. Before each session, the experimenter individually confirmed informed consent and photosensitivity-screening eligibility in person. The study also followed the withdrawal safeguards described in Section V-E, and only de-identified data are analyzed and reported.

ABSTRACT Screen content can be covertly photographed and transcribed by optical character recognition (OCR) or vision-language models (VLMs). We present temporal pixel masking, a display method exploiting differences between human temporal integration and short camera exposures. A base frame is decomposed into rapidly alternating, cryptographically randomized complementary subframes. We evaluate a 240-Hz software prototype using one USB Video Class (UVC) webcam. In the primary matched short-exposure analysis (12 content items $\times$ 9 geometries $\times$ 3 repeats), best-of-engine OCR character recovery decreased from 94.1% unprotected to 16.2% for the readability-priority profile and 5.0% for the high-suppression profile; paired content-cluster contrasts were 78.0% and 89.1%, respectively. The high-suppression result meets the descriptive $<$10% recovery target. Across the same nine geometries, 150-frame temporal averaging recovered 47.9–73.4% of characters for these two profiles. In the 0.5 m VLM probe, the strongest of three tested VLMs reached 77.8% exact match, with recovery concentrated on large-font short snippets. Under long exposure, masking could increase recovery over the unprotected baseline, consistent with reduced luminance shifting the sensor into its linear range. These results establish feasibility for mitigating conventional OCR in the tested short-exposure UVC-camera link, but not protection against temporal integration or VLM attackers. In 56 participants, readability-priority masking retained 94.83% typing accuracy but reduced speed by 1.14 words per minute (WPM); readability, stability, and comfort averaged 3.79, 2.98, and 3.39 out of 5, indicating measurable usability costs.

INDEX TERMS Adversarial noise, camera capture, display privacy, optical character recognition, persistence of vision, temporal masking, vision-language models, visual eavesdropping

I. INTRODUCTION

SCREENS routinely display credentials, financial records, source code, and other sensitive information in offices, banks, hospitals, and shared workspaces. At the same time, smartphones, smart glasses [1], and miniature cameras make physical screen capture increasingly unobtrusive. A single photograph can be processed by optical character recognition (OCR) or a vision-language model (VLM) without accessing the host device or leaving a network trace. Controlled experiments and user studies further show that visual eavesdropping can succeed quickly and often goes unnoticed [2], [3].

Existing defenses address different parts of this threat. Physical privacy filters and spatial display transformations restrict viewing geometry, while digital rights management and watermarking target digital copying or post-capture attribution [4], [5]. Temporal display techniques have also used persistence of vision and psychovisual modulation to separate what viewers and cameras perceive [6]–[10]. These studies establish temporal modulation as a viable display mechanism, but they leave an important empirical question unresolved. It remains unclear how much sensitive text is machine-recoverable from physical short-exposure captures

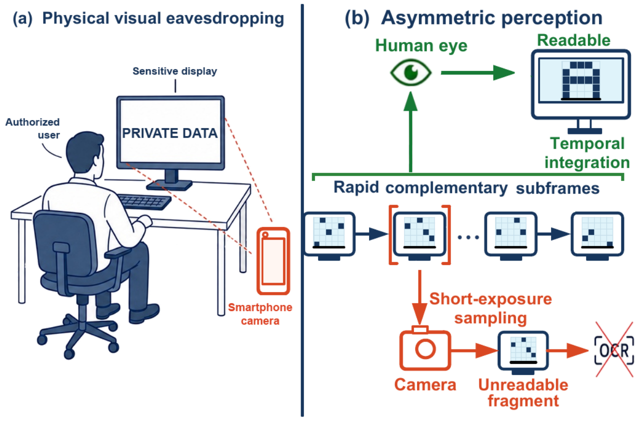

FIGURE 1. Threat scenario and operating principle of temporal pixel masking.

and how rapidly recovery increases with stronger recognition and temporal reconstruction methods.

We investigate this question through *temporal pixel masking*, a display-side implementation that builds on prior temporal masking principles. Each frame is decomposed into n rapidly alternating subframes whose pixel assignments are randomized, mutually exclusive, and collectively complete in the ideal digital domain. A human observer can integrate information across the display cycle, whereas a short camera exposure may intersect only part of that cycle and record incomplete character structure (Fig. 1). Profile-dependent stripe and glyph overlays, complementary noise, and a partial inversion frame provide additional operating points between readability and suppression. This mechanism does not prevent reconstruction in general: a long exposure or registered multi-frame sequence can accumulate the complementary information.

We implement the method as a 240-Hz graphics processing unit (GPU) rendering prototype and evaluate it with an instrumented USB Video Class (UVC) capture pipeline. The physical archive contains 10,575 captures from one webcam across three distances and three viewing angles. In the primary matched analysis at the common nominal exposure setting, conventional OCR character recovery decreases from 94.1% for the unprotected display to 16.2% for the readability-priority profile and 5.0% for the high-suppression profile. These results provide evidence of feasibility within the evaluated short-exposure camera link rather than a general guarantee against screen photography.

The evaluation therefore extends beyond favorable single-frame conditions. Fixed image preprocessing, long exposure, 150-frame temporal averaging, digital reconstruction, and VLM probes test how recovery changes as attacker capability increases. These attacks recover substantial information, especially from large-font short snippets, and expose the integration boundary of the approach. A 56-participant study separately measures the human side of the trade-off: readability-priority masking retains high typing accuracy but imposes measurable costs in speed, perceived stability, and visual comfort.

The main contributions are as follows:

1) **System-level physical evidence**: We provide a controlled measurement of short-exposure screen capture using archived physical images, multiple content types and geometries, three OCR engines, explicit sensitive-field recovery, matched profile comparisons, and content-cluster resampling. This evaluation quantifies temporal pixel masking as a system behavior rather than inferring protection from digitally generated subframes alone.
2) **Attack-boundary characterization**: We evaluate fixed preprocessing, long-exposure capture, temporal-average attacks, digital reconstruction, and commercially hosted VLMs. The resulting evidence identifies both the conditions under which conventional OCR recovery decreases and the conditions under which sustained capture or stronger recognition recovers substantial content. Detection and tracking experiments provide secondary cross-task stress tests.
3) **Security–usability trade-off**: We compare readability-priority and high-suppression operating points and conduct a controlled within-subject study with 56 participants. The results connect machine recovery to typing performance, readability, perceived stability, and immediate visual comfort, making the cost of the usability-facing configuration explicit.

The remainder of this paper reviews related work, defines the threat model and system design, and then presents the physical, attack-oriented, and usability evaluations. The final sections discuss the evidence boundaries and conclude the study.

II. RELATED WORK

A. SCREEN VISUAL EAVESDROPPING THREATS

Sensitive screen content can be acquired through direct photography, optical reflections, or non-visual side channels. Backes et al. [11] recovered displayed content from reflections on nearby objects, while Raguram et al. [12] reconstructed typed input from reflections and refractions in a user's surroundings. Electromagnetic attacks remove the need for an optical line of sight: Screen Gleaning recovered security codes from mobile-display emissions [13], and Deep-TEMPEST used learned reconstruction to recover text from unintended HDMI emissions [14]. Camera glasses further increase the likelihood of inconspicuous recording, although recent work by Wang et al. [1] studies wearer–bystander privacy expectations and protection preferences rather than the technical recovery of screen content.

Human-centered studies establish the practical relevance of these threats but do not by themselves characterize machine-readable leakage. An industry-sponsored visual-hacking experiment reported successful information capture in 91% of trials, with the first success occurring within 15 minutes in

49% of trials [2]. Eiband et al. [3] collected 174 accounts of shoulder-surfing incidents and found that observation was commonly opportunistic and often went unnoticed. These studies motivate display privacy, whereas the present work focuses on a narrower technical question: how much static text remains recoverable from short-exposure physical captures, and how that recovery changes under stronger acquisition and post-processing attacks.

B. DISPLAY PRIVACY PROTECTION

Existing defenses intervene at the camera, the display, or the surrounding illumination. ScreenAvoider detects displays and sensitive applications in images so that a cooperating capture or disclosure pipeline can suppress them [15], its protection therefore depends on control of the camera or downstream image handling. Display-side spatial methods instead alter what an observer sees. HideScreen suppresses low-frequency visual cues so that content blends into the background beyond a designed viewing distance [4]. EyeShield re-renders content to preserve close-range readability while reducing recognition at larger distances and viewing angles [16], in addition to user studies, it evaluated recognition on photographs using cloud OCR. Both systems primarily address distance- and angle-dependent shoulder surfing rather than temporally sampled display content. Illumination-based systems such as LiShield [17] and Rainbow [18] exploit rolling-shutter capture by modulating ambient light, but they protect illuminated physical scenes rather than altering the displayed pixels themselves.

Attribution mechanisms address a different stage of the leakage process. General image watermarking embeds identifiers that may survive subsequent processing [19], and screen-shooting-resilient schemes explicitly model the display–camera channel [5]. The mID system similarly creates camera-visible moiré identifiers for tracing leaked screen photographs [20]. These approaches can identify a leakage source after capture, but they are not designed to prevent an attacker from reading the photographed content. Visual cryptography is also conceptually related because it decomposes an image into shares that reveal the secret only when combined [21]. Its original threat model, however, concerns cryptographic share construction rather than time-varying display capture.

Temporal display techniques are the closest predecessors of the present work. Yerazunis and Carbone introduced time-masked privacy-enhanced displays [6]. Wu and Zhai subsequently formalized temporal psychovisual modulation [8], and Gao et al. applied that framework to an information-security display [9]. Lim used persistence of vision to construct browser keypads that resist a single digital screenshot [7]. Kaleido decomposes video into chrominance-complementary frames with controlled luminance variation so that viewers perceive the intended sequence while camera recordings are degraded [10]. Screen-camera communication explores the complementary objective: systems such as spatially adaptive embedding, DeepLight, and passive screen-to-camera communication encode camera-decodable information while limiting perceptual disturbance to human viewers [22]–[24].

These studies establish temporal modulation and multi-frame decomposition as prior mechanisms, the present work does not claim to introduce either concept. The unresolved issue is their security boundary for static sensitive text under modern machine recognition and increasingly capable capture strategies. Existing studies do not jointly measure short-exposure physical capture, multi-engine OCR, vision-language-model recovery, temporal integration, and phase-aware reconstruction. Our implementation also differs from Kaleido in mechanism: it uses randomized discrete pixel partitioning, and its exploratory high-suppression profile deliberately breaks strict per-cycle completeness (§IV-E). Table 1 compares representative systems by protection locus, threat model, and reported evaluation. Because the systems address different threats and evaluation targets, this is a contextual comparison rather than a matched-parameter performance ranking.

C. MACHINE RECOGNITION AND PHYSICAL-WORLD ROBUSTNESS

Research on adversarial examples provides a second point of comparison because it separates human readability from machine recognition. Song and Shmatikov showed that small digital modifications to text images can induce targeted OCR errors while preserving the apparent meaning for human readers [25]. More general digital attacks optimize model-specific perturbations [26]–[28], whereas physical attacks must remain effective after changes in viewpoint, illumination, printing, deformation, and camera processing [29]–[31]. Athalye et al. introduced Expectation over Transformation (EoT) to optimize adversarial examples over a chosen transformation distribution and demonstrated robust physical 3D objects [32]. Surveys of physical attacks and adversarial transferability show that robustness across capture conditions and generalization across models are distinct empirical questions [33], [34].

The display–camera channel adds transformations that are absent from ordinary digital evaluation. Niu et al. showed that simulated moiré patterns from screen photography can act as transferable adversarial perturbations [35]. Conversely, learned demoiréing can recover screen photographs degraded by such artifacts [36], illustrating why incidental optical distortion should not be treated as a stable security guarantee. The base mechanism studied here does not optimize a spatial perturbation against a particular recognizer, it partitions the displayed content over time. Enhanced profiles add noise and stripe or glyph overlays, but their effectiveness is measured after physical capture across multiple OCR and vision-language models rather than inferred from digital attack success. This distinction motivates our system-level evaluation of recovery and attack boundaries.

III. THREAT MODEL

TABLE 1. Scope comparison with representative screen-privacy systems.

Scheme	Protection locus	Target threat	Mechanism	Reported evidence and boundary
ScreenAvoider [15]	Camera or disclosure pipeline	Incidental capture of sensitive screens	Screen and application recognition, policy-based suppression	Lifelogging-image detection, requires a cooperating capture or disclosure path
HideScreen [4]	Mobile display	Human viewing beyond a designed distance	Spatial suppression of low-frequency cues	Human text/image recognition and usability across viewing distances
Eye-Shield [16]	Mobile display	Human viewing at larger distances and angles	Distance- and angle-aware spatial re-rendering	Human studies and photographed-content recognition, spatial rather than temporal defense
LiShield and Rainbow [17], [18]	Ambient illumination	Rolling-shutter photography of physical scenes	Temporally modulated LED illumination	Camera-image interference, requires controllable illumination and rolling-shutter interaction
Yerazunis and Gao [6], [9]	Displayed image	Incomplete temporal observation	Time masking or temporal psychovisual modulation	Proof-of-concept security displays, no modern OCR/VLM attack suite
Kaleido [10]	Displayed video	Continuous mobile-camera recording	Chrominance-complementary frames and luminance variation	PSNR, structural similarity, and subjective quality for protected video
This work	Static screen text	Short-exposure capture followed by machine recognition	Randomized pixel partitioning, high-suppression profile breaks completeness	OCR/VLM recovery, temporal integration, phase-aware reconstruction, and usability

A. THREAT SCENARIO AND PROTECTED ASSETS

The attacker is assumed to photograph the display from a direct or oblique angle and process the image, without compromising the host computer, application, or source document. Protected assets consist of confidential text-based information. The authorized user is expected to read and interact with the protected display directly. Our system does not prevent image acquisition, its narrower objective is to reduce machine-recoverable text in selected physical capture conditions while retaining usable visual content for the authorized user.

B. ATTACKER CAPABILITIES AND KNOWLEDGE

We organize the evaluated attacker along acquisition, temporal coverage, and recognition dimensions rather than as a strictly monotonic hierarchy. The *target attacker* obtains a short-exposure snapshot and applies conventional per-image OCR, this tier defines the primary feasibility claim. An *enhanced attacker* can vary viewing geometry, apply deterministic image preprocessing, select among recognition engines, or use a vision-language model (VLM). A *boundary attacker* can extend the exposure window, record a sustained sequence, align and aggregate multiple frames, or perform phase-aware reconstruction. The fixed preprocessing oracle covers the full short-exposure capture pool (§V-A), whereas the VLM probes are restricted to the high-suppression profile plus an unprotected original-short reference at on-axis geometries (§V-D). The shared capture, geometry, and preprocessing settings used for these tiers are specified in §V-A. The attack-specific configurations appear with the corresponding experiments, in particular the digital reconstruction attacks in §V-G2. The corresponding attack results are reported in §V-B–§V-G2.

We adopt a public-algorithm threat model: the attacker knows the temporal masking algorithm, subframe count n, refresh rate, profile composition, and candidate period parameters. Pixel-to-slot assignments and output permutations are randomized per cycle by a cryptographically secure pseudorandom number generator (CSPRNG; §IV). The attacker is not assumed to receive the current key or exact starting phase before capture, but neither value is treated as a security anchor. Temporal averaging can accumulate information without either value, and an adaptive attacker can estimate alignment from a captured sequence. Once registered observations cover a complete complementary cycle, key secrecy cannot prevent reconstruction in the ideal linear model.

C. SECURITY OBJECTIVES AND EXPLICIT BOUNDARIES

The primary confidentiality outcomes are OCR character recovery, whole-text exact match, and exact recovery of preidentified sensitive fields. The corresponding utility objective is direct human readability and interaction performance, assessed separately through typing behavior and subjective ratings. Under short exposure, the high-suppression profile uses descriptive targets of conventional-OCR character recovery $< 10\%$ and exact match $= 0\%$, the readability-priority profile targets a relative character-recovery reduction of at least 80% and exact match $< 1\%$. These thresholds organize the empirical comparison and are not formal security guarantees or device-independent deployment criteria. The high-suppression profile is an exploratory security-oriented stress point, whereas the user study evaluates the readability-priority profile.

The claimed protection scope excludes digital screenshots, host compromise, and source-document leakage. It also does not promise resistance to full-cycle exposure, registered multi-frame aggregation, phase-aware reconstruction, unconstrained learned restoration, or VLM-based inference. These stronger attacks are evaluated to identify where recovery increases or the protection fails, which is central to the attack-boundary contribution. Generalization beyond the tested display–camera link is not assumed. Object detection and tracking are reported only as secondary cross-task stress tests, while ΔE_{00} and the structural similarity index (SSIM) describe digital reconstruction quality rather than confidentiality.

IV. SYSTEM DESIGN

A. TEMPORAL SUBFRAME DECOMPOSITION

Given a normalized original frame $\mathbf{I} \in [0, 1]^{H \times W \times C}$, we generate binary masks $\mathbf{M}_1, \ldots, \mathbf{M}_n$ satisfying $\sum_{k=1}^{n} \mathbf{M}_k = \mathbf{1}$, and set $\mathbf{I}_k = \mathbf{I} \odot \mathbf{M}_k$. This yields:

- **Digital-domain completeness**: $\sum_{k=1}^{n} \mathbf{I}_k = \mathbf{I}$;
- **Mutual exclusivity**: each source-image pixel is assigned to exactly one of the n binary masks. Accordingly, its unmodified image contribution appears in exactly one base subframe.

The distinction between "summation" and "temporal averaging" matters: on an ideal linear display with equal slot durations and equal peak luminance, the average radiance over one cycle is $\mathbf{I}/n$, not $\mathbf{I}$, so digital-domain completeness means only that subframes can be re-summed, not that real display luminance has been losslessly preserved.

Pixel assignments are generated from a ChaCha20 [37] CSPRNG, with an HMAC-SHA256-derived subkey per cycle, rejection sampling mapping the byte stream to unbiased slot indices, and a separate Fisher–Yates shuffle [38] randomizing subframe playback order. A chi-square slot-count check ($\alpha = 0.01$, df $= n-1$, 3 at the evaluated $n = 4$), used only as an implementation sanity check with at most 5 resampling attempts, is not evidence of cryptographic strength. Per-cycle randomization avoids a fixed, periodically repeated pixel-to-slot pattern across cycles, but it cannot prevent linear reconstruction of a registered full cycle. This ChaCha20-based generator drives the Python physical-capture pipeline. The user-study web implementation reproduces the same mask structure, rejection sampling, and playback shuffling with a deterministic seeded 32-bit generator so that stimuli are exactly repeatable in the browser, but it has no CSPRNG properties (§V-E). Fig. 2 traces the source frame through complementary masks, optional noise and overlays, partial inversion, and high-refresh playback. It also contrasts the multi-slot integration available to human observers, long exposure, and video averaging with the sparse subframe that a short exposure may record.

B. COMPLEMENTARY ADVERSARIAL NOISE

We overlay adversarial noise $\mathbf{N}_1, \ldots, \mathbf{N}_n$ on the subframes, satisfying:

$$\sum_{k=1}^{n} \mathbf{N}_k = \mathbf{0} \tag{1}$$

Equation (1) holds strictly only in the ideal linear domain prior to clipping and the display transfer function. Motivated by prior work showing that small text-image perturbations can significantly mislead OCR systems [25], each cycle uses an available stored perturbation template when available, and otherwise a proxy-gradient perturbation generated by single-step fast gradient sign method (FGSM) [26] or iterative projected gradient descent (PGD) [27] with sign updates under an $\varepsilon = 8/255$ bound. The configured proxy set rotates across eight OCR and detector targets. Depending on availability, gradients come from the EasyOCR recognition network, a differentiable shadow objective, or a deterministic image-space surrogate. The resulting noise is then decomposed complementarily in the temporal domain. This mechanism is a practical transfer-oriented defense component, not a claim of white-box optimality for every recognizer or detector. The user-study web implementation substitutes a deterministic image-space texture of comparable amplitude for these model-gradient perturbations (§V-E). Because displays cannot emit negative radiance, the implementation uses a pedestal level and numerical clipping. Consequently, the noise is not guaranteed to cancel physically.

The components serve distinct functions in this system. The random dot-pattern mask is the primary temporal mechanism, while complementary noise is an optional ideal-domain component. Meanwhile, stripe/glyph overlays together with partial inversion are nonlinear additions used by the enhanced profiles below. The archived composite-profile definitions preserve correspondence with the captured data, and the physical analysis compares both individual ablations and complete profiles.

C. LUMINANCE MODEL ACROSS RENDERING PATHS

Because each pixel is illuminated only once per n-slot cycle, restoring the original cycle-averaged luminance in a linear-light model would require a factor-n light-output increase during the active slot. No evaluated path implements per-slot backlight control. The physical-capture playback fixes the composition gain at $\gamma = 1$ under fixed panel settings, leaving the linear-model cycle average at $\mathbf{I}/n$. The user-study web implementation instead applies a pixel-space gain of $1.1n$ with clipping to $[0, 255]$, which partially compensates the $1/n$ duty cycle but clips bright content. Its stimuli are therefore not photometrically identical to the physical-capture condition. Digitally integrated reconstructions are normalized before comparison, and CIEDE2000 ΔE_{00} [39] and SSIM [40] quantify these digital reconstructions only. None of these settings establishes radiometric equivalence on a real panel.

D. TIMING AND BANDWIDTH

For a basic cycle containing only n complementary subframes, taking 60 Hz as the minimum cycle rate requires $f_r \geq 60n$, at $n = 4$ this corresponds to 240 Hz. Including one inversion frame changes the requirement to $f_r \geq 60(n+1)$, dropping the actual cycle rate to only 48 Hz at 240 Hz. Thus, while 240 Hz meets the basic four-subframe configuration, it falls short of the inversion-frame configuration's 60 Hz cycle target. Because human flicker perception also depends on luminance, contrast, retinal eccentricity, spatial content, and eye movements [41], [42], nominal cycle rate alone cannot predict comfort.

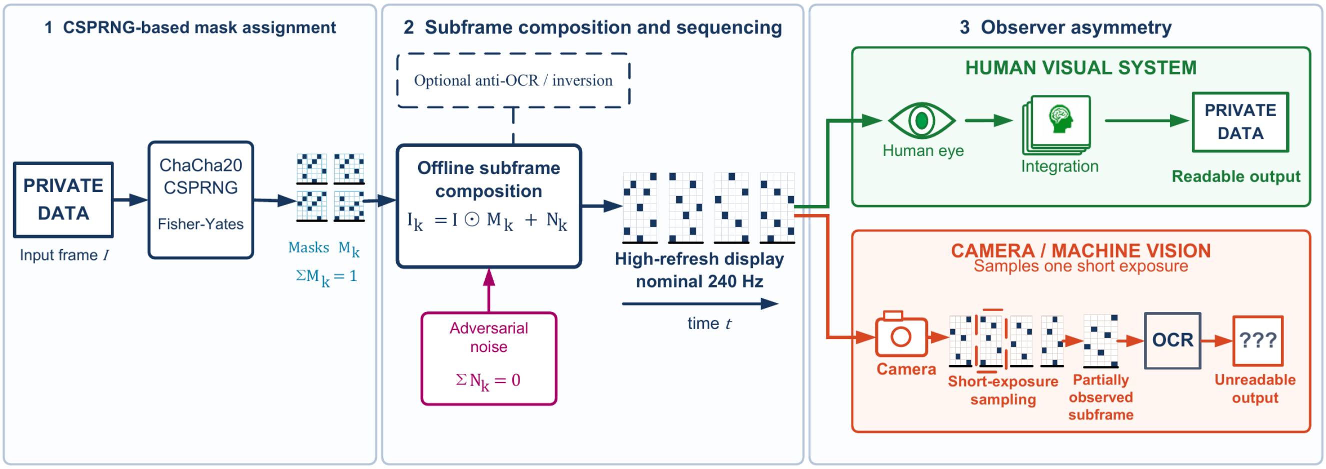

FIGURE 2. Temporal pixel masking system pipeline.

Whether an observer integrates complementary subframes into a perceptually complete image depends on the temporal contrast sensitivity function (TCSF) rather than on a single critical flicker fusion (CFF) cutoff. With high-frequency spatial edges, flicker can remain visible above 500 Hz [42]. The 48 Hz full-cycle rate of the inversion configuration may therefore produce visible flicker for some observers, the 60 Hz basic configuration cannot be assumed flicker-free, and the user study in §V-E evaluates readability and immediate comfort under these configurations.

E. OPERATING PROFILES AND CONFIGURABILITY

Our work is configurable, and the evaluation uses named, composable operating profiles rather than treating one parameter setting as universally appropriate. Unless otherwise stated, the protected profiles use $n = 4$ temporal frames, with complementary perturbation included as indicated in Table 2. The table defines the components of each profile, including the unprotected baseline. The exact parameter instances used in the physical-capture evaluation are reported separately in Section V-A. This separation distinguishes a reusable system interface from the particular configurations evaluated in this paper.

The nonlinear profile combinations are intentionally not interpreted as the sum of their isolated components. Compared to the readability-priority profile, the high-suppression profile uses coarser 2×2 pixel masks and denser stripes, significantly increasing stripe and glyph strengths. It also adds a dense line mesh and false strokes to aggressively degrade text recovery at the expense of readability. Because its visible artifacts may cause visual fatigue, this profile is unsuitable for default deployment and is included solely for experimental evaluation.

F. PARTIAL INVERSION FRAME

A long-exposure attack may cover a complete display cycle. To perturb, rather than eliminate, full-cycle accumulation, the evaluated offline-playback and web pipelines append a **partial inversion frame** $\alpha(\mathbf{1} - \mathbf{I})$ (normalized domain $\mathbf{I} \in [0, 1]$, 8-bit form $\alpha(255 - \mathbf{I}_8)$) after each n-subframe cycle, implemented as amplitude scaling over one additional display slot because vsync playback holds slot duration fixed. Higher α changes both the integrated signal and the digital reconstruction's luminance and contrast.

A real-capture ablation across $\alpha \in \{0, 0.2, 0.3, 0.5, 1.0\}$ (§V-C) informs this choice, the readability-priority setting of $\alpha = 0.2$ is a modeled design decision, not a validated optimum.

G. IMPLEMENTATION AND REPRODUCIBILITY

The proof of concept (PoC) operates at the application layer in Python (with NumPy, PyCryptodome, pygame/SDL, mss, ModernGL, etc.) and separates two rendering paths. One is the evaluated physical-capture path. It pre-generates all subframes offline with a CPU compositor and presents the sequence through pygame, requesting VSync and applying a software frame-rate cap when VSync is unavailable, no per-slot synthesis occurs during display. The other path is for user study. The user study runs a third, independent JavaScript/Canvas2D implementation (§V-E). Mask keys default to 32 bytes from the operating-system CSPRNG, and a fixed key can be supplied for regeneration. The playback metadata records permutations, the per-cycle noise schedule, and profile parameters, but not the key, so reproduction relies on the archived stimuli and explicit configurations rather than on command-level replay. No operating-system, display-driver, or backlight-level integration is implemented.

V. EXPERIMENTAL EVALUATION

A. EXPERIMENTAL SETUP

Hardware platform: All experiments used a desktop workstation with an Intel Core i7-8700K (3.70 GHz), 32 GB RAM, and an NVIDIA TITAN Xp (12 GB VRAM). The display was an AOC 24G51Z driven at a nominal 240 Hz. Each nominal

TABLE 2. Components of the evaluated operating profiles.

Profile	Temporal mask	Complementary noise	Additional display artifacts
Original (unprotected)	–	–	–
Mask only	✓	–	–
Mask + noise	✓	✓	–
Strong-overlay (anti-OCR)	✓	✓	Stripe and glyph overlays
Readability-priority	✓	✓	Stripe and glyph overlays with partial inversion
High-suppression	✓	✓	Enhanced overlays and partial inversion

240-Hz slot lasts 4.17 ms, while the basic and inversion-augmented cycles are 60 and 48 Hz.

Capture setup: All physical captures used one eMeet SmartCam S600 USB webcam with its standard UVC driver. Camera frames underwent perspective correction and content cropping before recognition. Per-capture metadata store exposure labels derived from DirectShow/UVC $\log_2$ controls.

The camera was placed at 0.5 m, 1 m, and 1.5 m from the display, rotated horizontally (in yaw) to $0°$, $15°$, and $30°$ at each distance, providing nine distance–angle configurations for the physical evaluation. The manufacturer specifies an 8-megapixel 4K sensor and 1080p/60 fps output [43].

All nine geometries used the same control values: -8 for short-exposure stills and video frames (nominally 3.91 ms), and -5 for long-exposure stills (31.25 ms). These values are control-derived labels. The display-brightness setting remained fixed, profiles were acquired in a fixed order, and video was captured nominally at 60 fps. Auto-exposure, gain, white balance, focus, and panel luminance were neither locked nor independently measured. Accordingly, the results characterize the evaluated display–camera link rather than an isolated temporal mechanism. The primary OCR inputs are geometrically corrected content crops.

Test content: Each position used 12 fixed items: two account strings, two code snippets, two digit strings, one English sentence, two mixed-language snippets, two English paragraphs, and one CET6 (College English Test Band 6) document. The CET6 page and two mixed-language items include Chinese characters. For short- and long-exposure stills, three repeats yield 36 captures per profile–attack–position cell and 324 across nine positions. For video, one 150-frame clip was planned per content–geometry cell, yielding 12 clips per position and 108 across nine positions; disclosed repeat-round clips are duplicate-averaged.

Evaluated configurations: Table 3 reports the exact profile instances used in the physical-capture evaluation. In particular, the physical-capture evaluation names the low-amplitude overlay instance `strong@overlay` (Strong-overlay): it uses the reusable `strong` profile with explicit stripe and glyph opacity overrides of 0.10 and 0.12, respectively. The readability-priority capture condition reuses this Strong-overlay instance and adds a partial inversion frame with $\alpha = 0.20$, whereas the reusable `strong` command-line default uses overlay amplitudes of 0.18 and 0.22. Relative to readability-priority, `high-suppression` further strengthens the masking configuration by increasing the cell size from 1 to 2 pixels, narrowing the stripe width from 10 to 6 pixels, and raising the stripe and glyph opacities from 0.10 and 0.12 to 0.42 and 0.55, respectively, while retaining the same partial inversion strength of $\alpha = 0.20$.

Evaluation metrics: Each captured image is processed independently by three OCR engines: Tesseract, EasyOCR, and Surya. OCR character recovery is defined as $\max(0, 1 - \mathrm{CER})$, where CER is character error rate. Exact match uses case-insensitive full-text comparison after whitespace removal. Explicit-field recovery uses 18 fields fixed before scoring across 9 field-bearing content items. A field counts as recovered only when its complete Unicode-normalized form appears in at least one engine output for that capture; case, whitespace, and punctuation are ignored. We report field-level micro recovery and sample-level macro recovery separately. Leak rate is the proportion of samples with character recovery $\geq 20\%$.

OCR aggregation and uncertainty: Character and whole-text metrics are maximized independently across the three engines for each capture. Explicit fields use the union of fields recovered by the three engines. Duplicate captures with the same profile, attack, content item, position, and repeat are averaged before summary or matching. Thus, the 324 units in each primary short-exposure profile comprise 12 content items, 9 geometries, and 3 repeats, which are not treated as 324 independent population samples. The 95% summary intervals reported in Table 6 resample these duplicate-averaged units with 2,000 resamples and seed 20260612. The primary comparison uses keys present in all three profiles; paired contrasts instead resample the 12 content items as clusters with the same seed and resample count, holding the tested geometries fixed. These intervals quantify sensitivity within the archived design.

Common-setting analysis: All 9 geometries define the primary fixed-setting result (Table 4). After duplicate averaging and three-profile matching, each profile contains $N = 324$ units. Table 6 reports the corresponding duplicate-averaged profile–attack summaries. Since duplicate captures are averaged within their unit, each planned content–geometry–repeat unit contributes once to these summaries.

Fixed preprocessing stress test: We evaluated Tesseract, EasyOCR, and Surya on all 1,107 primary short-exposure images spanning the 9 geometries. The fixed input grid comprised raw, gamma 0.5, luminance contrast-limited adaptive histogram equalization (CLAHE; 1st–99th percentile stretch, clip limit 4.0, 8×8 tiles), unsharp masking ($\sigma = 1.0$,

TABLE 3. Physical-capture profile instances for all masked profiles ($n = 4$).

Profile	Mask	Noise	Cell (px)	Stripe width (px)	Stripe α_s	Glyph α_g	Inv. α
Original (unprotected)	–	–	–	–	–	–	–
Mask only	✓	–	1	–	–	–	–
Mask + noise	✓	✓	1	–	–	–	–
Strong-overlay (anti-OCR)	✓	✓	1	10	0.10	0.12	–
Readability-priority	✓	✓	1	10	0.10	0.12	0.20
High-suppression	✓	✓	2	6	0.42	0.55	0.20

weights 1.7/−0.7), Gaussian adaptive thresholding (block 31, constant 5), and 2× bicubic upscaling. Parameters were fixed before scoring. For each metric and capture, the attacker retained the best engine–input pair; duplicate captures were then averaged before three-profile matching. The complete matrix contains 19,926 cells (1,107 images × 6 input forms × 3 engines) with no OCR errors. All cells used Tesseract 5.4.1 through `pytesseract 0.3.13`, EasyOCR 1.7.2, and Surya 0.14.7; raw and transformed inputs were evaluated on CUDA or CPU with the same software versions. The resulting oracle measures a reproducible fixed-grid preprocessing attack on the nine-geometry primary pool.

Digital reconstruction proxies: ΔE_{00} is the mean CIEDE2000 color difference between the original image and the full-cycle digitally reconstructed image; SSIM [40] is their mean structural similarity. Both metrics support within-paper comparison of digitally reconstructed profiles.

B. REAL-CAPTURE OCR EXPERIMENT

The physical archive comprises 10,575 images (1,175 per position), including ablations, parameter sweeps, and main attack conditions. Readability-priority includes a second capture round for 5 of 12 items, so the raw archive retains 459 short-exposure images and 153 videos across all positions rather than 324 and 108. These records are not discarded: duplicate keys are averaged within the matched content–geometry–repeat unit for Table 6 and the primary contrasts. Surya [44], EasyOCR [45], and Tesseract [46] produced 31,725 engine-level records. Each geometry uses an archived four-point homography and output canvas, followed by the recorded content crop. The primary OCR pipeline adds no adaptive enhancement. Profiles were collected in the order original, mask-only, mask+noise, Strong-overlay, readability-priority, and high-suppression. Fig. 3 shows digital reconstructions and matched captures at 1.0 m/0°.

Table 4 averages duplicates before matching, yielding $N = 324$ units per profile from 12 content items, 9 geometries, and 3 repeats. Paired contrasts resample the 12 content items as clusters while holding the 9 geometries fixed, so the 324 units are not treated as independent population samples. Contrasts use unrounded per-unit values and may differ by up to 0.1% from differences between the displayed rounded means.

For the field-level analysis in Table 5, 18 fields were fixed before scoring, and 243 of the 324 units per profile contain at least one field.

Table 6 reports duplicate-averaged best-of-engine results. Still rows use content–geometry–repeat units, whereas video rows use content–geometry units. Brackets are 95% unit-resampling summaries rather than population confidence intervals, and each video unit is the mean of 150 frames (nominally 2.5 s at 60 fps).

Key findings:

1) **Primary fixed-setting evidence**: Table 4 is the main short-exposure estimate. On the same 324 matched units per profile (12 content items × 9 geometries × 3 repeats), mean recovery is 94.1% unprotected, 16.2% for readability-priority, and 5.0% for high-suppression. The paired content-cluster contrasts are 78.0% ([75.5, 80.3]) and 89.1% ([85.0, 92.4]). Table 6 uses the same duplicate-averaged unit definition for its profile–attack summaries while the larger raw readability-priority capture counts remain disclosed for provenance.
2) **Fixed-grid preprocessing oracle**: Across the same 324 matched units per profile, raw-input recovery is 94.1% for unprotected content, 16.2% for readability-priority, and 5.0% for high-suppression. Selecting the metric-wise best output from the raw input and five fixed transforms across three OCR engines raises these values to 95.5%, 37.2%, and 13.2%, respectively (Table 8). The paired contrasts are 58.4% [53.4, 62.8] and 82.4% [78.1, 85.9], with 95% content-cluster bootstrap sensitivity intervals. Because selection uses the reference text, this oracle is a post hoc within-grid sensitivity analysis rather than a directly deployable attack or a general OCR upper bound. All 19,926 matrix cells completed without recorded runtime failures.
3) **Attacker-dependent recovery**: Mask-only, mask+noise, and Strong-overlay yield 19.2%, 25.6%, and 20.7% character recovery, respectively, showing nonmonotonic outcomes across these composite configurations but not isolating individual component effects. In the matched pool, readability-priority retains 15.7% field-micro exact recovery (Table 5). Its empirical P99 character recovery across the 324 duplicate-averaged best-of-three units is 94.8%, compared with 22.2% for high-suppression. Under raw-input OCR, both profiles meet their respective descriptive targets. Under the fixed-grid oracle, however, high-suppression rises to 13.2% character recovery and fails its joint target, while readability-priority reaches 37.2% recovery and 2.6% exact match, missing both of its targets. Target attainment therefore depends on the evaluated attack

TABLE 4. Primary matched short-exposure OCR comparison at the nominal 3.91-ms UVC setting.

Profile	N	Mean char recovery (%)	Unprotected − profile (%)	Interpretation
Original (unprotected)	324	94.1	–	Common-setting baseline
Readability-priority	324	16.2	78.0 [75.5, 80.3]	Conventional-OCR reduction, with upper-tail failures
High-suppression	324	5.0	89.1 [85.0, 92.4]	Exploratory stress profile; meets descriptive $<$ 10% target

TABLE 5. Exact recovery of preidentified sensitive fields in the matched pool.

Profile	Field micro (%; 486 opp.)	Sample macro (%)
Original (unprotected)	90.1	85.5
Readability-priority	15.7	15.6
High-suppression	1.0	1.0

setting.

4) **Recovery under integration attacks** (Fig. 4): Across the nine geometries, readability-priority reaches 60.9% char / 36.4% exact under nominal sensor-domain long exposure and 73.4% char / 43.5% exact under post-capture 150-frame averaging. Each video unit spans nominally 2.5 s at 60 fps. This single setting does not establish a duration–recovery curve. In the separate five-item ablation, weak inversion ($\alpha = 0.2$) yields only small point-estimate changes relative to $\alpha = 0$, without establishing equivalence. Mask-only and Strong-overlay yield close pooled estimates under 150-frame averaging (79.0% and 78.6%), but this descriptive comparison neither establishes equivalence nor isolates the overlay effect.
5) **Long-exposure reversal**: Under the common nominal 31.25 ms setting across nine geometries, character-recovery point estimates are 47.3% for unprotected content and 60.9% for readability-priority. Visible clipping occurs in some full-brightness captures. The pattern may reflect interactions among protected-image luminance, residual overlay edges, and geometry-dependent saturation or contrast compression; the unmeasured camera state prevents causal attribution. This reversal is confined to long exposure.
6) **High-suppression integration recovery**: This exploratory stress profile yields 9.3% char [7.9, 10.9] / 0% exact under nominal 31.25 ms long exposure and 47.9% char [41.8, 54.0] / 5.6% exact [1.9, 10.2] under the 150-frame mean. Brackets are 95% unit-resampling summaries pooled across nine geometries. Relative to readability-priority, high-suppression yields lower pooled recovery but does not prevent substantial multi-frame recovery. Its user acceptability was not evaluated.
7) **Geometry and repeat sensitivity**: Across all nine tested distance–angle geometries, character recovery for both readability-priority and high-suppression remains below the unprotected condition under short exposure. Averaging duplicate keys ensures that each planned content–geometry–repeat unit contributes once. In a descriptive sensitivity check, selecting round 1 or round 2 for the five repeated readability-priority items while retaining the other seven yields 16.6% and 15.8%, respectively, bracketing the reported duplicate-averaged estimate of 16.2%. Neither selection changes the profile ordering.

For each of the unprotected, readability-priority, and high-suppression profiles, Fig. 3 shows, from top to bottom, digital integration, a 3.91 ms capture, and a 31.25 ms capture. Camera panels are identical crops of archived raw JPEGs without geometric correction or brightness gain, whereas OCR uses geometrically corrected crops. Digital panels share a 1920×1080 black canvas and brightness-aligned full-cycle integration; unprotected integration equals the displayed source frame, and the protected reconstructions include the recorded $\alpha = 0.2$ inversion slot. These digital views do not measure human readability or visual comfort.

Fig. 4 presents best-of-engine character recovery and exact-match rates pooled across all nine geometries. Repeated captures are averaged within content–geometry–repeat units; readability-priority contributes $N = 324/324/108$ units for short exposure, long exposure, and video averaging. Error bars are unit-resampled 95% percentile-bootstrap intervals from 2,000 resamples, and lower values indicate less recovered text.

Per-engine character recovery: Table 7 reports descriptive raw-input point estimates after duplicate averaging. Unprotected recovery is 84.3% for Tesseract, 79.5% for EasyOCR, and 37.1% for Surya. Across engines, protected recovery ranges from 2.1–16.3% for Strong-overlay, 3.2–12.6% for readability-priority, and 1.8–2.0% for high-suppression. Because engine selection is performed per capture, the best-of-three result can exceed every single-engine mean. The reference-guided per-capture grid oracle in Table 8 and Fig. 5 yields 10.4%, 31.0%, and 18.3% recovery for readability-priority and 7.6%, 5.8%, and 6.1% for high-suppression using Tesseract, EasyOCR, and Surya, respectively. Because the raw input is included and selection uses the reference text, these increases quantify within-grid oracle gain rather than the performance of a single deployable transform.

Table 8 reports raw input $\rightarrow$ per-capture grid-oracle recovery for $N = 324$ matched units per profile after duplicate averaging. For the best-of-three oracle, whole-text exact match changes from 65.7% to 78.7% for Original, from 0.5% to 2.6% for readability-priority, and remains 0.0% for high-suppression.

Fig. 5 plots these within-engine increases. Connecting lines are not uncertainty intervals, and exact values appear in Ta-

TABLE 6. Duplicate-averaged real-capture OCR summary across nine geometries.

Profile	Attack mode	N	Char recovery (%) [95% summary]	Exact match (%)	Leak rate (%)
Original (unprotected)	Short exposure	324	94.1 [93.1, 95.0]	65.7	99.4
	Video (150-frame mean)	108	79.9 [73.5, 86.0]	61.1	86.1
	Long exposure	324	47.3 [42.8, 51.9]	18.8	58.6
Mask only	Short exposure	324	19.2 [17.0, 21.4]	0.3	30.6
	Video (150-frame mean)	108	79.0 [73.0, 84.9]	48.1	88.0
	Long exposure	324	67.2 [63.0, 71.5]	35.5	75.0
Mask + noise	Short exposure	324	25.6 [22.5, 28.8]	2.8	36.1
	Video (150-frame mean)	108	79.2 [73.3, 85.0]	49.1	88.9
	Long exposure	324	68.0 [63.7, 72.4]	39.8	76.2
Strong-overlay (anti-OCR)	Short exposure	324	20.7 [18.2, 23.3]	0.6	32.4
	Video (150-frame mean)	108	78.6 [72.5, 84.5]	50.9	88.9
	Long exposure	324	61.6 [57.1, 66.1]	39.8	71.3
Readability-priority	Short exposure	324	16.2 [14.1, 18.5]	0.5	23.5
	Video (150-frame mean)	108	73.4 [66.8, 79.6]	43.5	86.1
	Long exposure	324	60.9 [56.6, 65.6]	36.4	69.1
High-suppression	Short exposure	324	5.0 [4.3, 5.8]	0.0	3.7
	Video (150-frame mean)	108	47.9 [41.8, 54.0]	5.6	76.9
	Long exposure	324	9.3 [7.9, 10.9]	0.0	14.2

TABLE 7. Per-engine short-exposure OCR across nine geometries.

Profile	Engine	Char (%)	Exact (%)	N
Original	Surya	37.1	22.8	324
	EasyOCR	79.5	27.8	324
	Tesseract	84.3	47.8	324
Mask only	Surya	4.2	0.3	324
	EasyOCR	15.6	0.0	324
	Tesseract	3.2	0.0	324
Mask + noise	Surya	11.0	2.8	324
	EasyOCR	18.1	0.0	324
	Tesseract	4.8	0.0	324
Strong-overlay (anti-OCR)	Surya	6.1	0.6	324
	EasyOCR	16.3	0.0	324
	Tesseract	2.1	0.0	324
Readability-priority	Surya	3.6	0.5	324
	EasyOCR	12.6	0.0	324
	Tesseract	3.2	0.0	324
High-suppression	Surya	1.8	0.0	324
	EasyOCR	2.0	0.0	324
	Tesseract	1.9	0.0	324

TABLE 8. Character recovery under the fixed preprocessing grid.

	Character recovery (%)		
Engine	Original	Readability	High-supp.
Tesseract	84.3 → 91.4	3.2 → 10.4	1.9 → 7.6
EasyOCR	79.5 → 87.4	12.6 → 31.0	2.0 → 5.8
Surya	37.1 → 54.0	3.6 → 18.3	1.8 → 6.1
Best of three	94.1 → 95.5	16.2 → 37.2	5.0 → 13.2

ble 8.

C. INVERSION-FRAME ABLATION

A real-capture ablation (Table 9, pipeline per §V-B) compared a four-slot reference without an inversion frame ($\alpha = 0$) against five-slot cycles with an amplitude-scaled inversion frame ($\alpha > 0$). Thus, contrasts with $\alpha = 0$ do not isolate inversion strength because they also change the cycle length. The ablation holds the Strong-overlay instance fixed ($\alpha_s = 0.10$, $\alpha_g = 0.12$) and uses a five-item account/code/CET6 subset. Each setting contributes $N = 135$ long-exposure units and $N = 45$ video-averaging units from the five items across nine geometries, with three repeats for long exposure. Values are mean best-of-engine character recovery with 95% capture-resampling intervals. For $\alpha \leq 0.5$, recovery was nonmonotonic and remained 61.7–72.0% under long exposure and 64.3–72.7% under 150-frame averaging. At $\alpha = 0.2$, the corresponding estimates were 67.0% and 69.8%, versus 68.6% and 72.7% at $\alpha = 0$. These point estimates and intervals are descriptive, and interval overlap does not establish equivalence. Full inversion ($\alpha = 1.0$) reduced recovery to 18.1% and 28.4%, respectively, but visual utility was not evaluated across α. This subset differs from the 12-item readability-priority rows in Table 6; the $\alpha = 0.2$ long-exposure result is not a repeated estimate of that row. Accordingly, $\alpha = 0.2$ remains a modeled design choice rather than a validated optimum.

TABLE 9. Inversion-frame ablation on the fixed Strong-overlay profile.

α	Long-exp. char (%)	Video avg. char (%)
0.0	68.6 [63.1, 73.9]	72.7 [63.3, 81.1]
0.2	67.0 [61.3, 72.7]	69.8 [59.9, 78.7]
0.3	72.0 [66.2, 76.9]	64.3 [54.0, 73.6]
0.5	61.7 [55.6, 67.7]	66.4 [56.5, 75.2]
1.0	18.1 [14.0, 22.0]	28.4 [20.5, 36.9]

D. REAL-CAPTURE VLM PROBES

To assess the attack boundary against stronger recognizers, we evaluated three VLMs hosted through the SiliconFlow

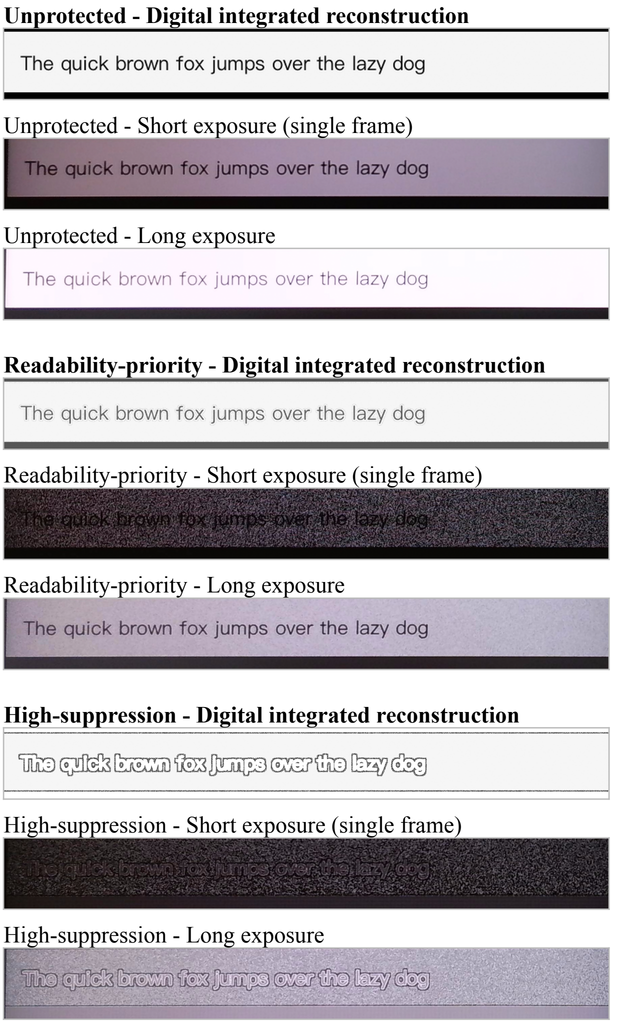

FIGURE 3. Matched digital reconstructions and real captures at 1.0 m and 0°.

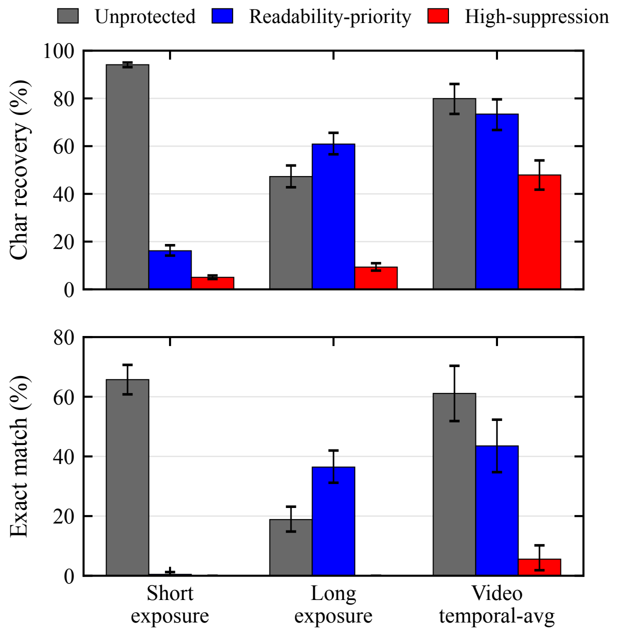

FIGURE 4. OCR recovery across profiles and capture attacks.

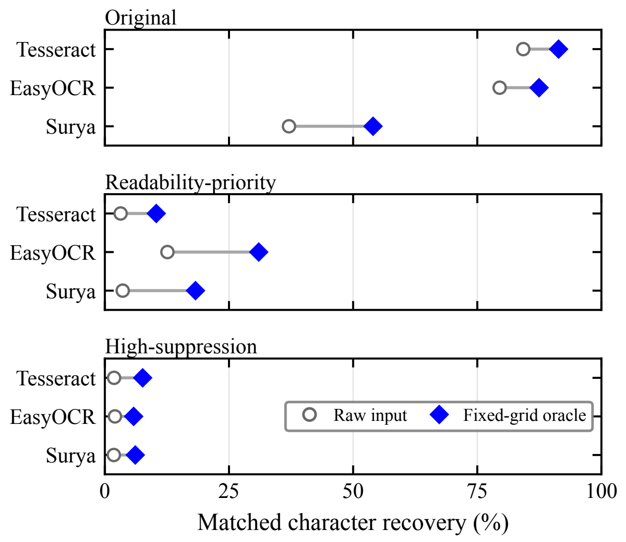

FIGURE 5. Per-engine OCR recovery before and after fixed-grid preprocessing.

API: Qwen3-VL-32B-Instruct [47], Kimi-K2.6 [48], and GLM-4.5V [49]. These probes measure text recovery from our captures rather than benchmark VLMs across platforms.

The probes used the same 12 content items at three on-axis positions (0.5, 1.0, and 1.5 m), a subset of the nine OCR geometries. Protected inputs used high-suppression, and the unprotected original-short reference was included only at 0.5 m. At each distance, the nominal 60 fps video data comprised 12 content-specific segments, each reduced to four views: *single best* (highest-contrast frame), *150-frame mean* (2.5 s), *window-mean best* (highest-luminance 30-frame mean), and *max projection* (temporal pixelwise maximum). The 0.5 m run contained 156 inputs: 36 unprotected short, 36 protected short, 36 protected long, and 48 video views. The 1.0 and 1.5 m runs each contained 120 protected inputs: 36 short, 36 long, and 48 video views. The VLMs and the OCR best-of-engine reference in Table 10 used the same file batch, nominal 3.91/31.25 ms labels, geometrically corrected crops, and unenhanced RGB inputs without photometric preprocessing. This matched external pipeline reduces image-preparation confounding in the OCR–VLM comparison.

All calls used SiliconFlow's OpenAI-compatible API. The recorded provider identifiers are `Qwen/Qwen3-VL-32B-Instruct`, `Pro/moonshotai/Kimi-K2.6`, and `zai-org/GLM-4.5V`. Qwen3-VL and GLM-4.5V have public technical reports [47], [49]. For Kimi-K2.6, the public model card [48], the provider endpoint identifier, and archived request metadata form the reproducibility record. Decoding temperature is 0, and the maximum output is 4,096 tokens.

The shared prompt requests only visible text and prohibits guessing; ground truth is not provided. Full prompts, parameters, and per-call outputs are archived. Metrics follow §V-A and are pooled at the capture level.

API outcomes were classified as parseable transcriptions, explicit refusals or valid empty outputs, and unresolved technical failures. Refusals and valid empty outputs received zero exact match and character recovery; other failures were retried up to three times and otherwise retained as missing. Each cell reports valid/planned calls and missing-response bounds obtained by assigning all unresolved calls either 0% or 100% recovery, and these bounds are not confidence intervals. Point entries indicate complete-return cells. Estimates conditional on successful calls were not used for model ranking, and direct cross-model comparisons were restricted to complete-return cells.

Of 1,188 planned requests, 1,133 (95.4%) returned valid outcomes, and 55 remained unresolved after three attempts. At 0.5 m, failures comprised two Qwen non-JSON responses, 31 GLM HTTP 429 errors, and one GLM non-JSON response. At 1.0 m, failures comprised four Qwen non-JSON responses, one Kimi connection closure, and 16 GLM TLS errors. No failures occurred at 1.5 m. The archived errors therefore indicate provider, transport, or formatting failures rather than semantic evidence of unreadability. Their endpoint- and session-clustering also prevents an assumption of random missingness.

We use the failure-free 1.5 m run as primary cross-model evidence. The 1.0 m run enters the short-exposure distance sweep as supplementary bounded evidence. At 1.0 m, long-exposure exact-match bounds are 75.0–86.1% for Qwen, 72.2–75.0% for Kimi, and 2.8–19.4% for GLM. Under the 150-frame mean, Qwen and Kimi each yield 83.3% exact match with 12/12 valid calls. The corresponding GLM bound is 58.3–75.0% with 10/12 valid calls. These incomplete cells are descriptive and are not used for direct model ranking. For compactness, Table 10 reports only the 150-frame mean at 0.5 m. The failure-free 1.5 m run reports all four aggregate views to show view-selection sensitivity.

The content breakdown in Table 11 uses same-session Qwen3-VL recovery on unprotected Original Short images as its baseline, not the OCR best-of-engine reference. Qwen and Kimi entries are complete-return point estimates; GLM entries give character-recovery bounds and valid/planned calls. In the failure-free 1.5 m run, CET6 dense pages yield 0% character recovery for all three models.

Table 10 presents the pooled VLM probe results, where short-exposure rows show distance variation while long-exposure and video aggregate rows expose integration attack boundaries. Table 11 breaks down 0.5 m high-suppression short-exposure character recovery by content type.

Key findings:

1) **VLMs recover substantially more text than conventional OCR**. In the 0.5 m probe, best-of-engine OCR yields 8.4% character recovery and 0% exact match on high-suppression short exposure. Qwen3-VL yields 91.5% character recovery and exactly recovers 28/36 captures. In the failure-free 1.5 m run, every endpoint returns 36/36 parseable outputs. Exact match ranges from 33.3% to 47.2%, and character recovery ranges from 61.5% to 70.0%.
2) **VLM recovery concentrates on large-font short snippets**. At 0.5 m, the failure-free CET6 cells yield 0.4–18.5% character recovery across the three models. The complete Qwen and Kimi cells yield 88.3–100% for credentials, code, and digit strings. GLM yields 88.9% for the complete code cell. Its missing-response bounds are 64.3–80.9% for credentials and 33.3–66.7% for digit strings. At 1.5 m, the failure-free CET6 cells yield 0% across all three models. The pattern is consistent with font scale and language priors [50], [51].
3) **Attack capability varies across models and conditions**. In the failure-free 1.5 m run, nominal long-exposure exact recovery ranges from 58.3% to 91.7%. Character recovery ranges from 62.4% to 89.9%. These complete cells show that endpoint, capture view, and content jointly shape the observed spread.
4) **Sustained video aggregation strongly increases VLM recovery**. At 0.5 m, Qwen and Kimi each yield 83.3% exact match under the 150-frame mean. Their character recovery is 94.0% and 95.5%, respectively. GLM has 8/12 valid calls, with bounds of 58.3–91.7% exact and 61.2–94.5% character recovery. In the failure-free 1.5 m run, the 150-frame mean yields 58.3–66.7% exact and 88.5–90.3% character recovery. Single-best-frame exact recovery is 0–8.3%. The corresponding 12.7–29.8% character values rest on low-information inputs. Language-prior completions may inflate them even though the prompt prohibits guessing, so exact match provides the more conservative reading.

E. USER EXPERIENCE STUDY

To evaluate the impact of the protected display on human readability and visual comfort, we designed and conducted a web-based user study.

Equipment and environment protocol: The same 240 Hz display as in §V-A. The formal study version sets a hard measured refresh-rate threshold of $\geq$200 Hz to ensure the fixed $n = 4$ basic subframe cycle rate $f_r/n \geq 50$ Hz, what below this threshold participation is blocked. Before each trial, display settings were standardized (fixed brightness, auto-brightness, etc.), background processes were closed, and viewing distance was approximate 60 cm.

The browser records actual frame intervals in each trial's `requestAnimationFrame` loop. If the interval between adjacent callbacks exceeds 1.5$\times$ the expected interval computed from the pre-trial measured refresh rate, a dropped-frame event is logged, along with estimated drop count, drop rate, mean/max frame intervals, observed refresh rate, basic subframe cycle rate, and full cycle rate with inversion frame.

Ethics and safety: This minimal-risk study involved in-person sessions with voluntary, screened participants. Before

TABLE 10. Real-capture VLM probes at 0° on-axis.

Distance (m)	Condition	Exact / character recovery (%)			
		OCR BoE	Qwen3-VL	Kimi-K2.6	GLM-4.5V
Short-exposure distance sweep					
0.5	original short	58.3/92.4	83.3/94.5 [36/36]	83.3/95.6 [36/36]	52.8–80.6/66.4–94.2 [26/36]
0.5	short	0.0/8.4	77.8/91.5 [36/36]	47.2/84.7 [36/36]	44.4–58.3/61.0–74.9 [31/36]
1.0	short	0.0/3.6	55.6/79.1 [36/36]	38.9/68.6 [36/36]	25.0–33.3/43.1–51.4 [33/36]
1.5	short	0.0/6.5	47.2/68.9 [36/36]	33.3/61.5 [36/36]	36.1/70.0 [36/36]
Key attack views					
0.5	long	0.0/2.1	77.8–83.3/80.9–86.5 [34/36]	2.8/7.9 [36/36]	0.0–16.7/4.1–20.8 [30/36]
0.5	Video (150-frame mean)	8.3/50.7	83.3/94.0 [12/12]	83.3/95.5 [12/12]	58.3–91.7/61.2–94.5 [8/12]
1.5	long	0.0/13.2	91.7/89.3 [36/36]	88.9/89.9 [36/36]	58.3/62.4 [36/36]
1.5	Video (single best)	0.0/4.3	8.3/24.8 [12/12]	0.0/12.7 [12/12]	0.0/29.8 [12/12]
1.5	Video (150-frame mean)	0.0/48.4	66.7/90.3 [12/12]	66.7/90.3 [12/12]	58.3/88.5 [12/12]
1.5	Video (window-mean best)	0.0/8.6	58.3/89.0 [12/12]	41.7/84.2 [12/12]	66.7/87.2 [12/12]
1.5	Video (max projection)	0.0/19.0	66.7/90.1 [12/12]	58.3/89.2 [12/12]	58.3/88.7 [12/12]

TABLE 11. High-suppression short-exposure VLM recovery by content type at 0.5 m.

Content type (planned)	Baseline (%)	Qwen (%)	Kimi (%)	GLM (%) [valid/planned]
CET6 document (3)	64.4	18.5	5.6	0.4 [3/3]
Credentials (6)	97.6	97.6	88.3	64.3–80.9 [5/6]
Code snippets (6)	89.8	94.9	95.8	88.9 [6/6]
Digit strings (6)	100.0	100.0	98.6	33.3–66.7 [4/6]
Mixed content (6)	100.0	100.0	92.5	65.5–98.8 [4/6]
Paragraphs (6)	100.0	99.8	81.7	66.5 [6/6]
English sentences (3)	95.3	95.3	96.9	94.6 [3/3]

each session, the experimenter obtained informed consent and confirmed photosensitivity safety. The webpage reiterated that the modulated display could cause discomfort and that participants could withdraw unconditionally at any time. Those with a history of photosensitive epilepsy or flicker sensitivity were excluded. While the implementation maintained a conservative operating floor, this did not constitute a clinical safety determination. At $n = 4$ on a 240 Hz panel the basic cycle rate is 60 Hz, and the inversion frame reduces the full cycle frequency to $f_r/(n+1) = 48$ Hz. Vision-correction status was collected to characterize the analyzed sample.

Task design: Participants completed a 12-second unmasked typing warm-up, a 10-second readability-priority preview, and an 8-second masked practice before four scored 20-second trials. On each typing screen, the stimulus was visible before the participant clicked "Start," with masked playback already active. Starting a trial initiated a 5-second countdown; input and timing began when the countdown ended.

Conditions and stimuli: Each participant completed two trials per condition: (A) an unmasked control ($n = 1$) and (B) readability-priority masking ($n = 4$, mask+noise, Strong-overlay with $\alpha_s = 0.10$ and $\alpha_g = 0.12$, and inversion with $\alpha = 0.2$). Trial order was counterbalanced using ABBA or BAAB. The masked renderer rotated through six distinct mask, noise, and stripe instances before repeating. Both conditions used the same Canvas rendering path, background, 23 px bold font, and color scheme. Their 220-character pseudo-word targets were independently generated by the same seeded procedure and matched in length and distribution, but not in literal content. The browser stimuli used the independent JavaScript/Canvas2D implementation described in §IV, rather than the Python physical-capture pipeline.

Scoring: Transcriptions were aligned with the optimal target prefix using Levenshtein minimum string distance (MSD) [52], yielding accuracy, characters per minute (CPM), and words per minute (WPM). First-keystroke latency was measured from the end of the countdown to the first nonempty input. Each participant's two trials per condition were averaged before paired analysis.

Subjective ratings: After typing, participants rated six display conditions on three 1–5 scales. Readability ranged from difficult to clear, stability from strong to no perceptible flicker, and visual comfort from very uncomfortable to very comfortable; higher scores indicated better outcomes. Conditions included the unmasked anchor, mask+noise at $n \in \{2, 3, 4\}$, $n = 4$ mask-only, and the $n = 4$ readability-priority configuration used for typing. The three mask+noise conditions operated at $\geq$ 60 Hz on the 240 Hz panel. A balanced Latin-square order counterbalanced position and carryover effects, and each condition was viewed for at least 10 seconds before rating. The high-suppression profile was not evaluated for acceptability.

Sample size and analysis plan: A minimum of $N = 24$ was planned for 80% power at paired effects of $d_z \approx 0.6$–0.8 and two complete 12-participant counterbalancing blocks. Pre-specified exclusions covered debug or incomplete sessions, refresh rates below 200 Hz, insufficient typing input, control accuracy below 50%, invalid temporal playback, and minimum-view straight-line ratings. The last criterion identified sessions in which all six ratings were submitted within 11 seconds and all three scores were identical within every condition. Typing outcomes were averaged by participant and condition, with speed interpreted jointly with accuracy. Paired differences used paired t-tests when the paired data passed the

TABLE 12. Within-subject typing outcomes for control and readability-priority masking ($N = 56$).

Metric	Control	Masked	Δ [95% CI]
Accuracy (%)	96.54	94.83	-1.71 [-3.39, -0.04]
WPM	20.44	19.30	-1.14 [-1.91, -0.39]
CPM	102.19	96.48	-5.71 [-9.56, -1.88]
Attempted chars	35.21	33.47	-1.74 [-2.88, -0.62]
First-key latency (ms)	742.64	833.61	$+90.97$ [-31.11, $+210.69$]

TABLE 13. Subjective ratings across six display conditions ($N = 56$; mean $\pm$ SD).

Condition	Readability	Stability	Comfort
Unmasked anchor	4.80 ± 0.64	4.79 ± 0.68	4.75 ± 0.61
n=2 mask+noise	4.61 ± 0.68	4.77 ± 0.57	4.48 ± 0.87
n=3 mask+noise	4.21 ± 0.87	4.57 ± 0.76	4.11 ± 0.91
n=4 mask+noise	4.23 ± 0.91	4.38 ± 0.93	4.02 ± 1.05
n=4 mask-only	4.16 ± 0.87	4.45 ± 0.78	4.02 ± 0.96
Readability-priority full (n=4)	3.79 ± 1.17	2.98 ± 1.30	3.39 ± 1.25

pre-specified Shapiro test; otherwise, Wilcoxon signed-rank tests were used. Effect sizes and participant-level bootstrap 95% confidence intervals were reported. Likert dimensions were analyzed using Friedman tests with Holm-corrected pairwise Wilcoxon tests.

Participants: Of 61 recruited volunteers, 56 were included in the analysis after applying the prespecified exclusion criteria. Five participants were excluded: three for minimum-view straight-line ratings and two for control mean accuracy below 50%. Among the analyzed participants, 46 wore glasses, one wore contact lenses, and nine reported no corrective lenses.

Table 12 uses participant means over two trials per condition, with Δ defined as masked minus control and participant-level bootstrap 95% confidence intervals. A paired t-test and Cohen's d_z are used when the pre-specified Shapiro test passes; otherwise, the analysis uses a Wilcoxon signed-rank test and rank-biserial r_{rb}. Test statistics are reported in the text.

Results: Table 12 and Fig. 6 summarize the user-study results. In Fig. 6, each gray line is one participant's mean across two trials per condition, and markers show group means with participant-level bootstrap 95% confidence intervals. The panels report transcription accuracy, words per minute, and first-keystroke latency. Readability-priority masking retained 94.83% mean accuracy but reduced WPM by 1.14 (95% CI [$-1.91, -0.39$], $d_z = -0.39$). After Holm correction, WPM, CPM, and attempted characters remained significantly lower, whereas accuracy and first-keystroke latency did not differ significantly.

All three subjective ratings varied across display conditions (Friedman tests, all $p < 10^{-13}$). Relative to the unmasked anchor, the readability-priority configuration received lower readability, stability, and comfort ratings, with large negative rank-biserial effects ($|r_{rb}| = 0.82$–0.97. All Holm-adjusted $p \leq 2.6 \times 10^{-4}$). Fig. 7 orders the six conditions from the unmasked anchor to the readability-priority configuration. Markers show means, and error bars show SD clipped to the

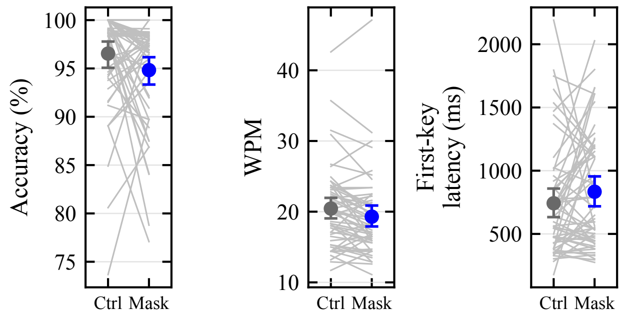

FIGURE 6. Within-subject typing performance under control and readability-priority masking.

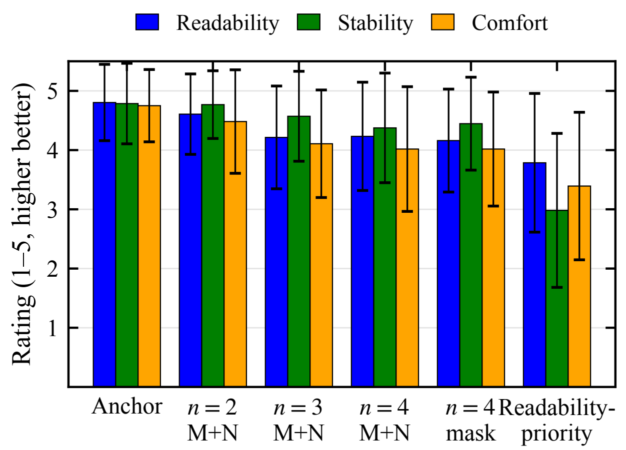

FIGURE 7. Readability, stability, and comfort ratings across six display conditions.

1–5 scale; higher values indicate better ratings.

F. REAL-CAPTURE OBJECT DETECTION AND TRACKING

To probe whether temporal degradation transfers beyond text, we report an exploratory detection/tracking stress test within the same display–camera link; its evidence strength is not equivalent to the matched OCR main experiment. Content was displayed on the 240 Hz screen, captured by the S600 at 1.5 m / 0°, and processed by four detectors [53]–[56]. mAP, mAP50, and average recall (AR) are reported for detection. Real-capture tracking uses higher-order tracking accuracy (HOTA) and identification F1 (IDF1); simulated tracking additionally reports multiple-object tracking accuracy (MOTA) and multiple-object tracking precision (MOTP). The protected condition uses the $n = 4$ mask+noise profile without overlays or inversion. The `real_clean` reference is an unprotected nominal 31.25-ms capture. The `real_short` image is the maximum-grayscale-contrast frame from a three-frame protected burst at the nominal 3.91-ms setting, and `real_video` is the mean of 150 protected frames acquired nominally at 60 fps. Tracking uses 450 frames from MOT17-09-FRCNN via still-display-then-capture and an in-project greedy association implementation

TABLE 14. Real-capture COCO detection across 150 images.

Detector	Condition	mAP (%)	mAP50 (%)	AR (%)
YOLO26x	real_clean	28.6	43.9	31.0
	real_short	1.3	2.4	1.4
	real_video	3.5	5.3	3.6
RT-DETR-x	real_clean	30.1	50.2	34.8
	real_short	3.5	5.8	4.1
	real_video	4.6	7.3	5.5
Faster R-CNN	real_clean	21.3	39.2	28.7
	real_short	0.8	1.6	0.9
	real_video	1.5	3.0	1.8
RetinaNet	real_clean	22.6	39.4	27.0
	real_short	0.9	1.8	1.0
	real_video	1.9	3.1	2.0

TABLE 15. Real-capture MOT17-09-FRCNN tracking across 450 frames.

Detector	Condition	HOTA (%)	IDF1 (%)
YOLO26x	real_clean	15.8	12.4
	real_short	8.7	6.5
	real_video	10.0	8.7
RT-DETR-x	real_clean	14.2	10.5
	real_short	8.1	5.4
	real_video	9.6	6.9
Faster R-CNN	real_clean	14.4	9.8
	real_short	6.8	6.2
	real_video	10.5	7.6
RetinaNet	real_clean	14.2	10.3
	real_short	5.1	4.4
	real_video	6.8	5.9

inspired by ByteTrack [57]; it does not establish performance for a continuous-video tracker.

Under short exposure, all four detectors' mAP50 drops from 39.2–50.2% to 1.6–5.8% (Table 14, Fig. 8), representing relative decreases exceeding 85%, while the `real_video` condition yields 3.0–7.3%, also below the `real_clean` baselines. Because protection, exposure, resolution, and frame reduction differ across conditions, these values are a cross-task stress test rather than an isolated protection effect. In still-frame tracking (Table 15), HOTA decreases from 14.2–15.8% to 5.1–8.7% and IDF1 from 9.8–12.4% to 4.4–6.5%. Detection mAP was computed by pycocotools. HOTA and IDF1 were both computed by TrackEval [58]. The tracker is an in-project greedy IoU association implementation.

G. SOFTWARE SIMULATION EXPERIMENTS

This section uses software simulation to address two questions: (1) what is the machine readability of one protected subframe obtained after the application renderer, and (2) what is recovery under zero camera noise and perfect alignment? By characterizing the post-render single-subframe capture point, the experiment defines a boundary that malware intercepting the original frame or accumulating a complete cycle would bypass.

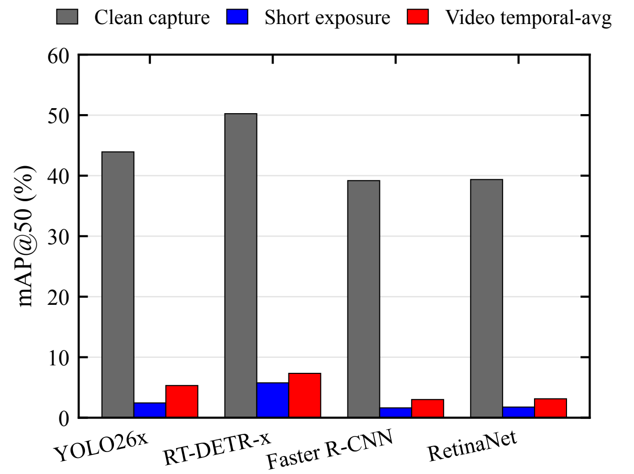

FIGURE 8. Real-capture mAP50 across clean, short-exposure, and video-mean conditions.

TABLE 16. Synthetic single-subframe OCR across three engines and 120 samples.

OCR engine	Original (%)	Single subframe (%)	Reduction (%)
Tesseract	94.0	0.0	94.0
EasyOCR	94.1	0.0	94.1
Surya	97.0	2.6	94.4

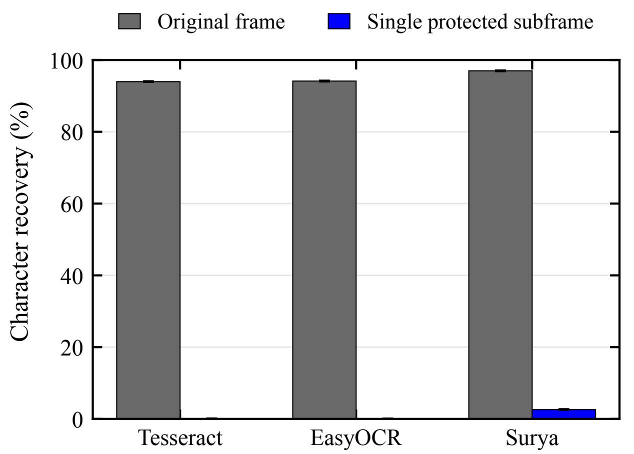

FIGURE 9. Character recovery on original frames and protected subframes.

1) Synthetic Single-Subframe OCR

Using Tesseract [46], EasyOCR [45], and Surya [44] on 120 synthetic samples, character recovery drops from 94.0–97.0% to 0–2.6% across three engines (Table 16). Protected subframes use $n = 4$ and $\varepsilon = 8/255$. Stratified breakdowns further characterize recovery across the evaluated content categories.

Fig. 9 visualizes these results with 95% bootstrap intervals resampled by corpus sample; single-subframe recovery does not exceed 2.6% for any engine.

TABLE 17. Archived digital reconstruction boundary for Tesseract.

Setting	N	Char. (%)	Exact (%)	SSIM	ΔE_{00}
Base: cycle mean	120	94.3	85.8	–	–
Base: per-sample strongest	120	95.2	91.7	–	–
Readability: cycle mean	16	87.9	50.0	0.948	1.46
High: cycle mean	16	56.6	6.3	0.764	4.87
High: inversion slot	16	93.0	81.2	–	–

2) Theoretical Security Boundary

When an attacker accumulates registered frames covering a full cycle, complementary subframes can be re-summed in the ideal linear model of (1), with the archived per-sample strongest digital attack recovering 95.2% and pure full-cycle temporal averaging recovering 94.3% (Table 17)—values that empirically mark the reconstruction boundary of the evaluated linear decomposition. Across the nine physical geometries, the high-suppression 150-frame mean yields 47.9% recovery. Character recovery also depends on content and the physical imaging path.

The 120-sample base attack and 16-sample profile ablation in Table 17 are separate archived digital evaluations. For each sample, "per-sample strongest" selects the best result among seven prespecified reconstructions: temporal average cycle, blue channel max, differential blue, differential luma, global shutter slot 0, phase search max, and phase search mean. The "inversion slot" row extracts only the $\alpha(\mathbf{1}-\mathbf{I})$ frame and applies contrast stretching; its 93.0% recovery identifies slot-level composition as a critical design variable.

The base rows are scored on the same 120-sample corpus as Table 16, whose unprotected Tesseract baseline is 94.0%. Reconstructions are normalized before scoring, so the 94.3% and 95.2% values fall marginally above that baseline. This margin reflects normalization and rounding rather than information gain; full-cycle reconstruction should therefore be interpreted as saturating at the unprotected level rather than exceeding it. Profile SSIM and ΔE_{00} describe full-cycle reconstruction quality, not physical or user-perceived quality.

3) Detection and Tracking Simulation

Simulations were run on COCO val2017 [59] and MOT17 [60] with YOLO26x [53], RT-DETR-x [54], Faster R-CNN [55], and RetinaNet [56]: the COCO simulation used 8 images as an implementation pipeline diagnostic (Table 18), while the MOT simulation covers 7 sequences totaling 5,316 frames and uses the same in-project ByteTrack-inspired greedy association method (Table 19). CLEAR metrics and IDF1 are computed by this in-project approximate implementation.

On the 8 COCO images, single-subframe mAP50 across the four detectors is 0–6.0%. Across the 5,316 MOT frames, single-subframe IDF1 is 0.1–5.4%, while MOTA includes negative and nonmonotonic values. Negative MOTA occurs when the false-positive, false-negative, and identity-error penalties exceed the number of ground-truth objects; it is not a negative detection probability. Detector architecture,

TABLE 18. Simulated COCO val2017 detection diagnostics across four detectors and eight images.

Detector	Condition	mAP (%)	mAP50 (%)	AR (%)
YOLO26x	clean	48.1	58.3	51.0
	single subframe	0.0	0.0	0.0
	temporal avg.	14.3	18.3	14.3
RT-DETR-x	clean	47.1	58.7	53.4
	single subframe	5.2	6.0	5.2
	temporal avg.	16.8	23.7	17.9
Faster R-CNN	clean	38.2	52.7	43.6
	single subframe	3.0	4.4	3.0
	temporal avg.	11.1	16.5	11.1
RetinaNet	clean	33.2	47.1	35.3
	single subframe	3.3	4.2	3.3
	temporal avg.	9.6	13.2	9.6

TABLE 19. Simulated MOT17 tracking across 5,316 frames.

Detector	Condition	MOTA (%)	MOTP (%)	IDF1 (%)
YOLO26x	clean	31.1	81.7	27.1
	single subframe	0.1	74.8	0.1
	temporal avg.	9.1	78.1	12.0
RT-DETR-x	clean	24.1	79.8	38.2
	single subframe	−2.3	73.2	5.4
	temporal avg.	−11.4	75.0	14.7
Faster R-CNN	clean	3.8	77.9	35.6
	single subframe	−2.5	71.2	4.5
	temporal avg.	−2.7	73.2	14.0
RetinaNet	clean	−5.4	77.1	30.5
	single subframe	0.4	70.6	2.7
	temporal avg.	−2.9	73.8	11.9

confidence thresholds, proposal density, and the greedy associator jointly shape the observed differences. Fig. 10 reports the largest model-wise protected-to-clean ratio within each evaluated task family: Surya for OCR and RT-DETR-x for detection and tracking. This conservative strongest-tested-pipeline summary uses task-specific metrics that are not directly commensurate.

VI. DISCUSSION

A. INTERPRETATION OF FINDINGS

The matched evaluation shows that temporal pixel masking substantially reduces conventional OCR recovery under the tested short-exposure conditions (Table 4). This result is specific to the evaluated display–camera link and does not establish general resistance to screen photography. The main contribution is therefore a system-level characterization of conditional effectiveness and attacker escalation. Detection and tracking remain secondary cross-task stress tests.

B. ATTACK BOUNDARIES AND TRADE-OFFS

Temporal masking relies on the mismatch between human integration and short camera exposure. However, a registered full cycle remains reconstructable, and physical frame averaging also recovers substantial text. The VLM probes further show that conventional OCR underestimates stronger recog-

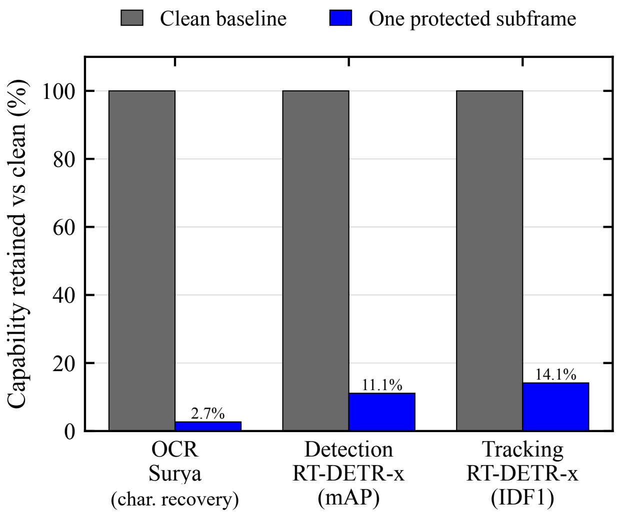

FIGURE 10. Strongest tested protected-to-clean ratio within each task family.

nition attackers. Protection claims must therefore specify the attacker, capture mode, and content regime.

The profiles expose a security–utility trade-off. Readability-priority preserves greater reconstruction fidelity but leaves more residual text, whereas high-suppression improves resistance with greater distortion. These digital metrics do not directly represent human experience. The user study nevertheless indicates that readability-priority preserves typing accuracy while reducing speed, stability, and comfort.

Kaleido uses complementary color modulation to preserve the intended visual appearance [10]. Our enhanced profiles, by contrast, do not require exact full-cycle reconstruction. Recovery from the inversion slot nevertheless shows that relaxing this constraint does not eliminate reconstruction risk. Temporal masking should therefore complement field-level policies and independent access controls.

C. LIMITATIONS AND FUTURE WORK

The physical evidence is limited to one webcam and a fixed profile order. Exposure values are control-derived labels, while camera state, ambient illumination, display–camera phase, slot timing, and panel luminance remain unmeasured. These coupled factors limit causal isolation and external generalization.

The corpus and tested geometries cannot support broader script- or device-level inference. The aggregation experiment covers one duration, and detection and tracking remain exploratory stress tests. The short browser-based user study does not evaluate high-suppression.

Future work should use randomized capture, locked camera parameters, photometric controls, additional devices, duration sweeps, and longer cross-profile usability studies.

VII. CONCLUSION

We evaluated temporal pixel masking as a display-side mitigation against camera-based text recovery. Across nine geometries in one UVC display–camera link, the matched short-exposure analysis showed substantially lower conventional OCR recovery. Mean best-of-engine character recovery decreased from 94.1% unprotected to 16.2% for readability-priority and 5.0% for high-suppression. However, 150-frame averaging recovered 73.4% and 47.9%, respectively. The fixed-grid preprocessing oracle, digital reconstruction, and VLM probes exposed additional recovery paths, limiting protection claims to the evaluated attacker and capture conditions.

In a short browser-based study with 56 participants, readability-priority masking retained 94.83% typing accuracy but reduced speed by 1.14 WPM. Lower subjective stability and comfort ratings revealed additional usability costs. Together, these findings provide system-level evidence of conditional effectiveness, usability trade-offs, and attack boundaries. Temporal pixel masking can mitigate conventional short-exposure OCR in the tested setting, but does not provide a general guarantee against sustained capture or stronger recognition.

ETHICS STATEMENT

This research is defensive in nature and evaluates recovery boundaries to assess protection against unauthorized screen photography, not to provide an attack guide. The user study involved minimal-risk behavioral research. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type, so no institutional determination was available. Before each session, the experimenter individually confirmed voluntary consent and photosensitivity-screening eligibility in person; the web interface recorded the confirmation state and timestamp. The protocol included in-person supervision and voluntary withdrawal as described in §V-E, and analysis and publication use only de-identified data.

DATA AVAILABILITY

The source code, experimental configurations, analysis pipelines, and curated benchmark data supporting the findings of this study are available at https://github.com/docandyc/temporal-pixel-masking.

The code is distributed under the MIT license. Full raw capture datasets and raw participant databases are withheld due to volume and privacy considerations; these materials are available from the corresponding author upon reasonable request.

REFERENCES

[1] X. Wang, K. Peng, X. Yi, and H. Li, "Mind the gap: Mapping wearer-bystander privacy tensions and context-adaptive pathways for camera glasses," in *Proc. CHI Conf. Human Factors in Computing Systems*, 2026, pp. 1–28, doi: 10.1145/3772318.3791848. [Online]. Available: https://doi.org/10.1145/3772318.3791848

[2] Ponemon Institute, "Global visual hacking experiment," 3M, Tech. Rep., 2016, Accessed: Jul. 2, 2026. [Online]. Available: https://www.3m.com/3M/en_US/privacy-screen-protectors-us/expertise/visualhacking/
[3] M. Eiband, M. Khamis, E. von Zezschwitz, H. Hussmann, and F. Alt, "Understanding shoulder surfing in the wild: Stories from users and observers," in *Proc. CHI Conf. Human Factors in Computing Systems*, 2017, pp. 4254–4265, doi: 10.1145/3025453.3025636.
[4] C.-Y. Chen, B.-Y. Lin, J. Wang, and K. G. Shin, "Keep Others from Peeking at Your Mobile Device Screen!" in *Proc. 25th Annu. Int. Conf. Mobile Computing and Networking (MobiCom)*, 2019, pp. 1–16, doi: 10.1145/3300061.3300119.
[5] H. Fang, W. Zhang, H. Zhou, H. Cui, and N. Yu, "Screen-shooting resilient watermarking," *IEEE Trans. Information Forensics and Security*, vol. 14, no. 6, pp. 1403–1418, 2019.
[6] W. S. Yerazunis and M. Carbone, "Privacy-enhanced displays by time-masking images," in *Proc. Australian Conf. Computer-Human Interaction (OzCHI)*, Nov. 2001. [Online]. Available: https://www.merl.com/publications/TR2002-11
[7] J. Lim, "Defeat spyware with anti-screen capture technology using visual persistence," in *Proc. 3rd Symp. Usable Privacy and Security (SOUPS)*, 2007, pp. 147–148, doi: 10.1145/1280680.1280701.
[8] X. Wu and G. Zhai, "Temporal psychovisual modulation: A new paradigm of information display," *IEEE Signal Process. Mag.*, vol. 30, no. 1, pp. 136–141, 2013, doi: 10.1109/MSP.2012.2219678.
[9] Z. Gao, G. Zhai, and X. Min, "Information security display system based on temporal psychovisual modulation," in *Proc. IEEE Int. Symp. Circuits and Systems (ISCAS)*, 2014, pp. 449–452, doi: 10.1109/ISCAS.2014.6865167.
[10] L. Zhang, C. Bo, J. Hou, X.-Y. Li, Y. Wang, K. Liu, and Y. Liu, "Kaleido: You can watch it but cannot record it," in *Proc. 21st Annu. Int. Conf. Mobile Computing and Networking (MobiCom)*, 2015, pp. 372–385, doi: 10.1145/2789168.2790106.
[11] M. Backes, T. Chen, M. Dürmuth, H. P. A. Lensch, and M. Welk, "Tempest in a teapot: Compromising reflections revisited," in *Proc. IEEE Symp. Security and Privacy (S&P)*, 2009, pp. 315–327, doi: 10.1109/SP.2009.20.
[12] R. Raguram, A. M. White, D. Goswami, F. Monrose, and J.-M. Frahm, "iSpy: Automatic reconstruction of typed input from compromising reflections," in *Proc. 18th ACM Conf. Computer and Communications Security (CCS)*, 2011, pp. 527–536, doi: 10.1145/2046707.2046769.
[13] Z. Liu, N. Samwel, L. Weissbart, Z. Zhao, D. Lauret, L. Batina, and M. Larson, "Screen Gleaning: A screen reading TEMPEST attack on mobile devices exploiting an electromagnetic side channel," in *Proc. Network and Distributed System Security Symp. (NDSS)*, 2021. [Online]. Available: https://arxiv.org/abs/2011.09877
[14] S. Fernández, E. Martínez, G. Varela, P. Musé, and F. Larroca, "Deep-TEMPEST: Using deep learning to eavesdrop on HDMI from its unintended electromagnetic emanations," in *Proc. 13th Latin-American Symp. Dependable and Secure Computing (LADC)*, 2024, pp. 91–100, doi: 10.1145/3697090.3697094.
[15] M. Korayem, R. Templeman, D. Chen, D. J. Crandall, and A. Kapadia, "ScreenAvoider: Protecting computer screens from ubiquitous cameras," *arXiv preprint arXiv:1412.0008*, 2014, doi: 10.48550/arXiv.1412.0008. [Online]. Available: https://arxiv.org/abs/1412.0008
[16] B. Tang and K. G. Shin, "Eye-Shield: Real-time protection of mobile device screen information from shoulder surfing," in *Proc. 32nd USENIX Security Symp.*, 2023, pp. 5449–5466. [Online]. Available: https://www.usenix.org/conference/usenixsecurity23/presentation/tang
[17] S. Zhu, C. Zhang, and X. Zhang, "Automating visual privacy protection using a smart LED," in *Proc. 23rd ACM Int. Conf. Mobile Computing and Networking (MobiCom)*, 2017, pp. 329–342, doi: 10.1145/3117811.3117820.
[18] L. Yang, W. Wang, Z. Wang, and Q. Zhang, "Rainbow: Preventing mobile-camera-based piracy in the physical world," in *Proc. IEEE INFOCOM*, 2018, pp. 1061–1069, doi: 10.1109/INFOCOM.2018.8485881.
[19] X. Zhong, A. Das, F. Alrasheedi, and A. Tanvir, "A brief, in-depth survey of deep learning-based image watermarking," *Applied Sciences*, vol. 13, no. 21, p. 11852, 2023, doi: 10.3390/app132111852.
[20] Y. Cheng, X. Ji, L. Wang, Q. Pang, Y.-C. Chen, and W. Xu, "mID: Tracing screen photos via moiré patterns," in *Proc. 30th USENIX Security Symp.*, 2021, pp. 2969–2986. [Online]. Available: https://www.usenix.org/conference/usenixsecurity21/presentation/cheng-yushi
[21] M. Naor and A. Shamir, "Visual cryptography," in *Proc. EUROCRYPT*, 1994, pp. 1–12.
[22] V. Nguyen, Y. Tang, A. Ashok, M. Gruteser, K. Dana, W. Hu, E. Wengrowski, and N. Mandayam, "High-rate flicker-free screen-camera communication with spatially adaptive embedding," in *Proc. IEEE INFOCOM*, 2016, pp. 1–9, doi: 10.1109/INFOCOM.2016.7524512.
[23] V. Tran, G. Jayatilaka, A. Ashok, and A. Misra, "DeepLight: Robust & unobtrusive real-time screen-camera communication for real-world displays," in *Proc. 20th ACM/IEEE Int. Conf. Information Processing in Sensor Networks (IPSN)*, 2021, pp. 238–253, doi: 10.1145/3412382.3458269.
[24] S. K. Ghiasi, M. Kaldenbach, and M. Zuniga, "Passive screen-to-camera communication," in *Proc. 20th Int. Conf. Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)*, 2024, pp. 35–43, doi: 10.1109/DCOSS-IoT61029.2024.00016.
[25] C. Song and V. Shmatikov, "Fooling OCR systems with adversarial text images," *arXiv preprint arXiv:1802.05385*, 2018, doi: 10.48550/arXiv.1802.05385. [Online]. Available: https://arxiv.org/abs/1802.05385
[26] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2015.
[27] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2018.
[28] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Security and Privacy (S&P)*, 2017, pp. 39–57.
[29] K. Eykholt, I. Evtimov, E. Fernandes, B. Li, A. Rahmati, C. Xiao, A. Prakash, T. Kohno, and D. Song, "Robust physical-world attacks on deep learning visual classification," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 1625–1634.
[30] S. Thys, W. Van Ranst, and T. Goedeme, "Fooling automated surveillance cameras: Adversarial patches to attack person detection," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019.
[31] K. Xu, G. Zhang, S. Liu, Q. Fan, M. Sun, H. Chen, P.-Y. Chen, Y. Wang, and X. Lin, "Adversarial T-shirt! Evading person detectors in a physical world," in *Proc. European Conf. Computer Vision (ECCV)*, 2020, pp. 665–681.
[32] A. Athalye, L. Engstrom, A. Ilyas, and K. Kwok, "Synthesizing robust adversarial examples," in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 284–293.
[33] H. Wei, H. Tang, X. Jia, Z. Wang, H. Yu, Z. Li, S. Satoh, L. Van Gool, and Z. Wang, "Physical adversarial attack meets computer vision: A decade survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 9797–9817, 2024.
[34] J. Gu, X. Jia, P. de Jorge, W. Yu, X. Liu, A. Ma, Y. Xun, A. Hu, A. Khakzar, Z. Li, X. Cao, and P. Torr, "A survey on transferability of adversarial examples across deep neural networks," *Transactions on Machine Learning Research*, 2024, [Online]. Available: https://openreview.net/forum?id=AYJ3m7BocI.
[35] D. Niu, R. Guo, and Y. Wang, "Moiré attack (MA): A new potential risk of screen photos," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 26 117–26 129.
[36] B. Liu, X. Shu, and X. Wu, "Demoiréing of camera-captured screen images using deep convolutional neural network," *arXiv preprint arXiv:1804.03809*, 2018, doi: 10.48550/arXiv.1804.03809. [Online]. Available: https://arxiv.org/abs/1804.03809
[37] D. J. Bernstein, "ChaCha, a variant of Salsa20," in *Workshop Record of SASC*, vol. 8, 2008, pp. 3–5.
[38] D. E. Knuth, *The Art of Computer Programming, Volume 2: Seminumerical Algorithms*, 3rd ed. Addison-Wesley, 1997.
[39] G. Sharma, W. Wu, and E. N. Dalal, "The CIEDE2000 color-difference formula: Implementation notes, supplementary test data, and mathematical observations," *Color Research & Application*, vol. 30, no. 1, pp. 21–30, 2005.
[40] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, 2004.
[41] Y. Cai, A. Bozorgian, M. Ashraf, R. Wanat, and R. K. Mantiuk, "elaTCSF: A temporal contrast sensitivity function for flicker detection and modeling variable refresh rate flicker," in *Proc. SIGGRAPH Asia*, 2024.
[42] J. Davis, Y.-H. Hsieh, and H.-C. Lee, "Humans perceive flicker artifacts at 500 Hz," *Scientific Reports*, vol. 5, 2015, Art. no. 7861, doi: 10.1038/srep07861.

[43] EMEET, "SmartCam S600: 4K Ultra HD webcam," 2026, Accessed: Jul. 7, 2026. [Online]. Available: https://emeet.com/en-eu/products/webcam-s600

[44] V. Paruchuri, "Surya: Multilingual document OCR toolkit," 2024. [Online]. Available: https://github.com/VikParuchuri/surya

[45] JaidedAI, "EasyOCR: Ready-to-use OCR with 80+ supported languages," 2020. [Online]. Available: https://github.com/JaidedAI/EasyOCR

[46] R. Smith, "An overview of the Tesseract OCR engine," in *Proc. Int. Conf. Document Analysis and Recognition (ICDAR)*, 2007, pp. 629–633.

[47] S. Bai *et al.*, "Qwen3-VL Technical Report," *arXiv preprint arXiv:2511.21631*, 2025, doi: 10.48550/arXiv.2511.21631. [Online]. Available: https://arxiv.org/abs/2511.21631

[48] Moonshot AI, "Kimi-K2.6 model card," Hugging Face, 2026, accessed: Jul. 11, 2026. [Online]. Available: https://huggingface.co/moonshotai/Kimi-K2.6

[49] GLM-V Team, W. Hong, W. Yu, X. Gu *et al.*, "GLM-4.5V and GLM-4.1V-Thinking: Towards Versatile Multimodal Reasoning with Scalable Reinforcement Learning," *arXiv preprint arXiv:2507.01006*, 2025, doi: 10.48550/arXiv.2507.01006. [Online]. Available: https://arxiv.org/abs/2507.01006

[50] Y. Shi, D. Peng, W. Liao, Z. Lin, X. Chen, C. Liu, Y. Zhang, and L. Jin, "Exploring OCR capabilities of GPT-4V(ision): A quantitative and in-depth evaluation," *arXiv preprint arXiv:2310.16809*, 2023, doi: 10.48550/arXiv.2310.16809. [Online]. Available: https://arxiv.org/abs/2310.16809

[51] C. Jiang, Z. Wang, M. Dong, and J. Gui, "Survey of adversarial robustness in multimodal large language models," *arXiv preprint arXiv:2503.13962*, 2025, doi: 10.48550/arXiv.2503.13962. [Online]. Available: https://arxiv.org/abs/2503.13962

[52] R. W. Soukoreff and I. S. MacKenzie, "Metrics for text entry research: An evaluation of MSD and KSPC, and a new unified error metric," in *Proc. SIGCHI Conf. Human Factors in Computing Systems (CHI)*, 2003, pp. 113–120, doi: 10.1145/642611.642632.

[53] G. Jocher, J. Qiu, M. Liu, S. Lyu, F. C. Akyon, and M. E. Kalfaoglu, "Ultralytics YOLO26: Unified real-time end-to-end vision models," 2026, arXiv:2606.03748, doi: 10.48550/arXiv.2606.03748. [Online]. Available: https://arxiv.org/abs/2606.03748

[54] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 16 965–16 974.

[55] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 28, 2015, pp. 91–99.

[56] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2980–2988.

[57] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-object tracking by associating every detection box," in *Proc. European Conf. Computer Vision (ECCV)*, 2022, pp. 1–21.

[58] J. Luiten, A. Ošep, P. Dendorfer, P. Torr, A. Geiger, L. Leal-Taixé, and B. Leibe, "HOTA: A higher order metric for evaluating multi-object tracking," *Int. J. Comput. Vis.*, vol. 129, no. 2, pp. 548–578, 2021, doi: 10.1007/s11263-020-01375-2.

[59] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Proc. European Conf. Computer Vision (ECCV)*, 2014, pp. 740–755.

[60] A. Milan, L. Leal-Taixé, I. Reid, S. Roth, and K. Schindler, "MOT16: A benchmark for multi-object tracking," *arXiv preprint arXiv:1603.00831*, 2016.

ZHEYUAN HUANG is currently pursuing the B.S. degree in software engineering with Zhejiang University of Water Resources and Electric Power. His current work focuses on computer vision and artificial intelligence.

CHEN XING received the M.Sc. degree in software engineering of distributed system from the KTH Royal Institute of Technology, Stockholm, Sweden. He is currently a Lecturer with the Zhejiang University of Water Resources and Electric Power, Hangzhou, China. His current research interests include computer graphics and neural networks.